Dimitris Papantzikos
ML Engineer

papantzikos12@gmail.com
+45 91 62 87 19
Copenhagen, Denmark
github.com/tzikos
Personal Website

ML engineer focused on applying AI to practical problems in healthcare and energy. I break work down to first principles and have hands-on experience across the lifecycle: building ML pipelines, automating documentation with LLMs, and deploying production systems. Currently pursuing an M.Sc. in Mathematical Modelling and Computation at DTU; motivated by projects where AI delivers real and visible impact, (in)directly enhancing people's lives.

Professional Experience

Present – Aug 2025 | **Student Worker (AI/ML), Modelling & Optimisation, Vattenfall, Copenhagen, Denmark**
- Automated technical documentation workflows using LLM-powered assistants.
- Developed internal AI chatbots for non-technical stakeholders.
- Applied supervised learning to physics-based energy optimisation problems.

LLMs | retrieval-augmented generation | supervised learning | wake modelling

Present – Jul 2025 | **AI Software Developer, Schellinger Capital / Kapital&Connect, Copenhagen, Denmark**
- Build ML-driven systems for automated investor–startup matching.
- Deliver a user-friendly web application.

Machine learning | matching algorithms | web applications

Aug 2025 – Sep 2023 | **Data & Research Analyst, Recognyte, Remote**
- Built ETL pipelines and automated reporting systems across Python, PostgreSQL and Tableau.
- Developed NLP pipelines and ML-assisted property valuation models.
- Created Python-based web scraping automation tools.

Python | PostgreSQL | Tableau | BI automation | NLP | web scraping

Aug 2023 – Nov 2022 | **Data Analyst, Data to Action, Athens, Greece**
- Implemented forecasting models for retail analytics using scikit-learn.
- Built SQL-based data acquisition pipelines and Tableau dashboards.
- Developed web scraping solutions for market data collection.

scikit-learn | SQL pipelines | Tableau | web scraping | forecasting

Education

2024 – 2026 M.Sc. Mathematical Modelling and Computation, Technical University of Denmark (DTU), Copenhagen, Denmark — focus on Machine Learning, Deep Learning and programming.

2016 – 2021 B.Sc. Mathematics, Aristotle University of Thessaloniki, Thessaloniki, Greece — focus on Data Analysis.

Projects

Mamba architectures for EEG signal abnormality detection (Academic project) 2025

Benchmarked Mamba models for detecting abnormalities in EEG signals.

Mamba models | EEG | benchmarking

Machine Learning Operations | Plant leaf health classification (Course project) 01/2025

Trained and deployed a plant leaf health model on Google Cloud using Vertex AI for training, Cloud Run for deployment and Cloud Build for Docker builds; FastAPI for the backend, Streamlit for the frontend, version control with git/GitHub and CI/CD with GitHub Actions.

Google Cloud (Vertex AI | Cloud Run | Cloud Build) | FastAPI | Streamlit | git | GitHub Actions

GradeAid — AI-assisted learning platform (Personal project) 2025

Built a full-stack learning assistant using PostgreSQL, LangChain, and OpenAI APIs.

PostgreSQL | LangChain | OpenAI API | full-stack

Automated job scraper & Discord bot (Personal automation) 2025

LinkedIn scraping pipeline with near-real-time Discord notifications.

Web scraping | automation | Discord bots

Copenhagen Apartments | BoligPortal Rental Price Prediction (Course project) 2024

Built a classical neural network with PyTorch to predict rental prices from apartment features, achieving a mean absolute error of 2000 kroner.

PyTorch | neural networks | regression | model evaluation

Patient mortality classification with Electronic Health Records (Course project) 11/2024

Re-developed the EHRMamba model for the Physionet2012 dataset with PyTorch, leveraging DTU GPU resources to reach ~85% accuracy on early patient mortality prevention.

PyTorch | deep learning | EHRMamba | GPU training

Graphical LASSO implementation (Academic project) 2024

Implemented sparse inverse covariance estimation using NumPy, CVXPY, and Matplotlib.

NumPy | CVXPY | Matplotlib | graphical models

HPC thermal simulation (Academic project) 2024

GPU-accelerated heat diffusion simulation using CUDA, Numba, and CuPy.

CUDA | Numba | CuPy | GPU acceleration

Copenhagen apartment finder (Personal project) 2024

Web scraping and analytics platform built with Streamlit.

Streamlit | web scraping | analytics

Personal Tableau projects (Portfolio) 2023–2024

Collection of Tableau dashboards exploring diverse datasets and storytelling angles.

Tableau | data storytelling

Awards & Talks

- 1st place, Valhacks hackathon (BEST x IDA with Accenture, 2025) for a feature-engineered matching solution using FAISS vector stores.
- Top 4% in Data Art & Storytelling Competition (Data2Speak / Alex Kolokolov, 2024) with a finalist dashboard.
- Speaker, Athens Tableau User Group (03/2024) on data visualization practices and personal Tableau projects.

Skills

Python	pandas, numpy, scikit-learn, pytorch, selenium, fastapi, pytest, streamlit, beautifulsoup
LLMs & NLP	LangChain, retrieval-augmented generation, OpenAI API, chatbot development
SQL	PostgreSQL, database management & monitoring, pgAdmin4
Tableau	Server management, visualizations, data sources
Data & Ops	Linux bash scripting; Git, GitHub, GitHub Actions; CI/CD; Google Cloud (Vertex AI, Cloud Run, Cloud Build)
R	Multivariate statistics, PCA, CCA, visualizations
Other	Microsoft SSIS

Volunteering & Community

- Speaker, Athens Tableau User Group (03/2024) — Presented data visualization practices and personal Tableau projects.
- Mathematics Tutor, Metadrasi (05/2023 – 05/2024, Athens, Greece) — Tutored refugee children aged 12–16 in mathematics.

Interests

- Photography — Social events, product and portrait photography.
- Sports — Strength training, running, martial arts (kickboxing, judo), marathon runner, triathlete.
- Hiking and camping.